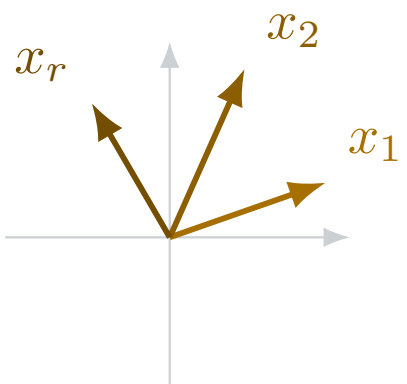
given basis

$x_1, \ldots, x_r$ span V



linearly independent,
not necessarily orthogonal

Gram-Schmidt

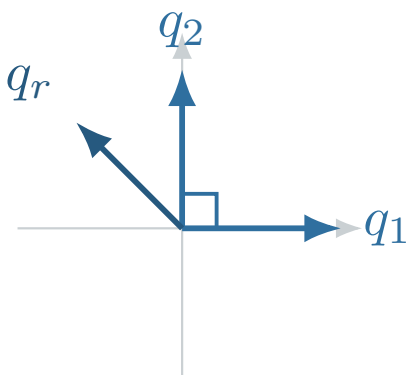
$$u_k = x_k - \sum_{j<k} \langle x_k, q_j \rangle q_j$$

$$q_k = \frac{u_k}{\|u_k\|}$$

remove earlier directions, then normalize

orthonormal basis

$q_1, \ldots, q_r$ span the same V



$\langle q_i, q_j \rangle = 0 \ (i \neq j)$

$\|q_i\| = 1$

$$\text{span}\{x_1, \ldots, x_k\} = \text{span}\{q_1, \ldots, q_k\}$$

$$Q^\mathsf{T} Q = I_r$$